

Lab05

AUTHOR

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leitura dos pacotes

```
library(tidyverse)
```

— Attaching core tidyverse packages — tidyverse 2.0.0 —

✓ dplyr 1.2.1 ✓ readr 2.2.0

✓ forcats 1.0.1 ✓ stringr 1.6.0

✓ ggplot2 4.0.3 ✓ tibble 3.3.1

✓ lubridate 1.9.5 ✓ tidyr 1.3.2

✓ purrr 1.2.2

— Conflicts — tidyverse_conflicts() —

✗ dplyr::filter() masks stats::filter()

✗ dplyr::lag() masks stats::lag()

! Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors

```
library(leaflet)
```

```
library(readr)
```

leitura do banco de dados flights

```
analise_aeronave <- function(tail_number) {  
  
  dados_aeronave <- list()  
  
  teste <- function(data, pos) {  
  
    bloco <- data %>%  
      filter(TAIL_NUMBER == tail_number)  
  
    if (nrow(bloco) > 0) {  
      dados_aeronave[[length(dados_aeronave) + 1]] <- bloco  
    }  
  }  
  
  read_csv_chunked(  
    file.path("dados", "flights.csv"),  
    callback = DataFrameCallback$new(teste),  
    chunk_size = 100000  
  )  
  
  dados_aeronave <- bind_rows(dados_aeronave)  
  
  return(dados_aeronave)  
}
```

```
dados_aeronave <- analise_aeronave("N431WN")
```

— Column specification —

```
cols(
  .default = col_double(),
  AIRLINE = col_character(),
  TAIL_NUMBER = col_character(),
  ORIGIN_AIRPORT = col_character(),
  DESTINATION_AIRPORT = col_character(),
  SCHEDULED_DEPARTURE = col_character(),
  DEPARTURE_TIME = col_character(),
  WHEELS_OFF = col_character(),
  WHEELS_ON = col_character(),
  SCHEDULED_ARRIVAL = col_character(),
  ARRIVAL_TIME = col_character(),
  CANCELLATION_REASON = col_character()
)
i Use `spec()` for the full column specifications.
```

filtrar

```
dados_aeronave <- dados_aeronave %>%
  arrange(YEAR, MONTH, DAY, DEPARTURE_TIME)
```

leitura do banco de dados airport

```
airports <- read_csv(file.path("dados", "airports.csv"))
```

Rows: 322 Columns: 7

— Column specification —

```
Delimiter: ","
chr (5): IATA_CODE, AIRPORT, CITY, STATE, COUNTRY
dbl (2): LATITUDE, LONGITUDE
```

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show\_col\_types = FALSE` to quiet this message.

```
names(airports)
```

```
[1] "IATA_CODE" "AIRPORT"   "CITY"      "STATE"     "COUNTRY"   "LATITUDE"
[7] "LONGITUDE"
```

juntar as coordenadas de origem

```
dados_geo <- dados_aeronave %>% left_join(airports %>% select(IATA_CODE, LATITUDE, LONGITUDE))
dados_geo <- dados_geo %>% left_join(airports %>% select(IATA_CODE, LATITUDE, LONGITUDE)) %>% r
```

calcular velocidade média

```
dados_geo <- dados_geo %>% mutate(velocidade_media = if_else(!is.na(AIR_TIME) & AIR_TIME>0, DI
```

criar mapa interativo

```
mapa <- leaflet() %>%
  addTiles()

mapa <- mapa %>%
  addCircleMarkers(
    lng = dados_geo$ORIGIN_LON,
    lat = dados_geo$ORIGIN_LAT,
    popup = dados_geo$ORIGIN_AIRPORT,
    radius = 5
  )
```

Warning in validateCoords(lng, lat, funcName): Data contains 156 rows with either missing or invalid lat/lon values and will be ignored

```
for (i in 1:nrow(dados_geo)) {

  mapa <- mapa %>%
    addPolylines(
      lng = c(
        dados_geo$ORIGIN_LON[i],
        dados_geo$DESTINATION_LON[i]
      ),
      lat = c(
        dados_geo$ORIGIN_LAT[i],
        dados_geo$DESTINATION_LAT[i]
      ),
      weight = dados_geo$velocidade_media[i] / 50,
      popup = paste(
        "Origem:", dados_geo$ORIGIN_AIRPORT[i],
        "<br>Destino:", dados_geo$DESTINATION_AIRPORT[i],
        "<br>Velocidade média:",
        round(dados_geo$velocidade_media[i], 1),
        "mph"
      )
    )
}

list(
  tabela = dados_geo,
  mapa = mapa
)
```

\$tabela

A tibble: 1,938 × 36

	YEAR	MONTH	DAY	DAY_OF_WEEK	AIRLINE	FLIGHT_NUMBER	TAIL_NUMBER
	<dbl>	<dbl>	<dbl>	<dbl>	<chr>	<dbl>	<chr>
1	2015	1	1	4	WN	1762	N431WN

2	2015	1	1	4 WN	3564 N431WN
3	2015	1	1	4 WN	3564 N431WN
4	2015	1	1	4 WN	3564 N431WN
5	2015	1	2	5 WN	1793 N431WN
6	2015	1	2	5 WN	1793 N431WN
7	2015	1	2	5 WN	1793 N431WN
8	2015	1	2	5 WN	1793 N431WN
9	2015	1	2	5 WN	1793 N431WN
10	2015	1	2	5 WN	3144 N431WN

i 1,928 more rows

i 29 more variables: ORIGIN_AIRPORT <chr>, DESTINATION_AIRPORT <chr>,
SCHEDULED_DEPARTURE <chr>, DEPARTURE_TIME <chr>, DEPARTURE_DELAY <dbl>,
TAXI_OUT <dbl>, WHEELS_OFF <chr>, SCHEDULED_TIME <dbl>, ELAPSED_TIME <dbl>,
AIR_TIME <dbl>, DISTANCE <dbl>, WHEELS_ON <chr>, TAXI_IN <dbl>,
SCHEDULED_ARRIVAL <chr>, ARRIVAL_TIME <chr>, ARRIVAL_DELAY <dbl>,
DIVERTED <dbl>, CANCELLED <dbl>, CANCELLATION_REASON <chr>, ...

\$mapa

```
Sys.setenv(TZ = "America/Sao_Paulo")  
Sys.time()
```

[1] "2026-09-17 00:10:21 -03"